

Longlong Xu

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Class of 2028 | Full-Time & Internship Target Roles: LLM Post-Training; Multimodal LLMs; Coding Agents

Education

Tsinghua University | Department of Computer Science and Technology | *Ph.D. Candidate* Sep. 2023–Jun. 2028
Advisor: Prof. Dan Pei; Research interests: LLM post-training, multimodal LLM alignment and reasoning, Agentic RL, and applications to AIOps, including anomaly detection and root-cause analysis.

Tsinghua University | Department of Computer Science and Technology | *B.Eng.* Sep. 2018–Jun. 2023
GPA: 3.88/4.00 (32/207)

Internship Experience

LLM-Focused Internships

ByteDance | *Research Intern* Sep. 2025–Aug. 2026
Built multimodal LLM post-training and evaluation pipelines for reasoning over time-series data, covering controllable data synthesis, multi-turn tool-use trajectory construction, SFT, Agentic RL, and trajectory evaluation. Extended LLaMA-Factory, VeRL, TRL GRPOTrainer, and vLLM to support training and serving for native new-modality LLMs.

- **ChatTS – Native Time-Series Multimodal LLM**: Contributed to the first LLM that natively accepts time series as a new modality; aligned the modality through SFT and integrated it into vLLM for efficient inference. Published at **VLDB 2025 (CCF A; 4th author, 2nd student author)**; 470+ GitHub stars, 120+ citations, and 20K+ model downloads as of Aug. 2026.
- **FoundRoot – Structured Reasoning for Root-Cause Analysis**: Organized time-series semantics, anomalous fluctuations, and fault-propagation relationships into structured reasoning chains; improved generalization and interpretability on unseen fault types with GRPO + RLVR. Published at **ICSE 2026 (CCF A; co-author, 2nd student author)**.
- **IntentAD – User-Intent-Aligned KPI Anomaly Detection Agent**: Built a multi-step agentic decision process spanning intent parsing, memory retrieval, and tool execution, and injected the capability through Agentic RL. Deployed in ByteDance's development environment; first-author submission to **ICSE 2027 (CCF A)**.
- **G2AD – VLM-Based Active Perception for Time-Series Anomaly Detection**: Built a multi-turn inspection agent on Qwen3.5-9B that screens candidates globally, gathers high-resolution local evidence in parallel, refines anomaly boundaries across turns, and stops adaptively. Trained with controllable data synthesis and two-stage SFT/GRPO; first-author submission to **AAAI 2027 (CCF A)**.

ZTE | *Research Intern* Jan. 2024–Aug. 2024
Built an end-to-end workflow for domain corpus and QA synthesis, LLM post-training, and automatic evaluation in AIOps; improved domain QA accuracy by more than 80% over the chat base model.

- **OpsLLM**: Designed multi-task QA synthesis from technical documents; evaluated CPT, SFT (LoRA/full-parameter), and DPO, mitigated catastrophic forgetting through mixed general-domain data and warmup, and used RAFT-based multi-hop data for RAG-enhanced SFT.
- **OpsEval**: Co-built an evaluation suite with 9,000+ QA pairs across eight AIOps subdomains and multiple evaluation paradigms, and designed FAE-Score to align automatic scoring with expert judgment. Published at **FSE 2025 (CCF A; 3rd author, 2nd student author)**; the leaderboard has received 2M+ visits.
- **TechSupportEval**: Combined key-term matching, step-order verification, and completeness checks to improve evaluation AUC on TechQA by 7.6%. Published at **IJCNN 2025 (THU B / CCF C; 4th author, 3rd student author)**.

Additional Research Internships

ByteDance | *Research Intern* Jan. 2025–Aug. 2025
Studied robustness and inference efficiency for time-series foundation models, addressing shortcut reliance on recent observations and limited access to ultra-long contexts; produced **two first-author ICML 2026 papers (CCF A)**.

Huawei | *Research Intern* Sep. 2024–Dec. 2025
Researched multisource spatio-temporal foundation models for optical transport networks and automatic evaluation-data generation from hierarchical documents; produced **one co-authored FSE 2026 paper (CCF A)** and supported engineering deployment.

eBay | *Research Intern* Aug. 2023–Dec. 2023
Developed root-cause localization for large-scale microservice systems, resulting in **one co-authored KDD 2024 paper (CCF A)** and an engineered deployment.

RealAI | *Research Intern* Aug. 2022–Jul. 2023
Studied physical-world 3D adversarial attacks on face recognition, producing a **3rd-author CVPR 2023 Highlight paper** and a **2nd-author IJCV 2024 paper** (both CCF A); the methods informed internal defense research.

Academic Research

Off-Policy Optimization for LLM Post-Training and Agentic RL | SIS, IPD

Ongoing

- Contributed to Selective Importance Sampling (SIS), addressing off-policy correction in LLM RL post-training. The method applies token-level rejection sampling, treats accepted tokens as on-policy, and retains standard importance-sampling correction for rejected tokens; planned for submission to ICLR 2027 as 3rd author.
- Proposed Inherited Policy Drift (IPD) for long-horizon Agentic RL, where later updates inherit policy mismatch accumulated along prior decisions. The method estimates update reliability from current-token drift and cumulative drift over past agent actions, then continuously reweights the policy loss to suppress stale-trajectory interference; planned for first-author submission to ICLR 2027.

LLM Post-Training for Anomaly Detection | IntentAD, G2AD

Aug. 2025–Present

- Built an end-to-end pipeline spanning controllable trajectory synthesis, SFT, Agentic RL, multi-turn tool use, and evaluation, targeting user-intent alignment and fine-grained anomaly localization over time series.
- Proposed IntentAD, an agentic decision process for intent parsing, memory retrieval, and tool execution; proposed G2AD, a VLM-based active-perception loop that alternates global screening with parallel local inspection and iteratively refines anomaly boundaries. First-author submissions to ICSE 2027 and AAAI 2027, respectively.

Multimodal LLM Alignment and Reasoning over Time Series | ChatTS, FoundRoot Jan. 2025–Jul. 2025

- Contributed to ChatTS (VLDB 2025, CCF A; 4th author, 2nd student author), enabling SFT-based alignment between time-series data and language space and adapting training and serving frameworks for a native new-modality LLM.
- Contributed to FoundRoot (ICSE 2026, CCF A; co-author, 2nd student author), organizing multimodal semantics, anomalous fluctuations, and propagation relationships into structured root-cause reasoning chains and improving generalization through GRPO + RLVR.

LLM Evaluation for AIOps | OpsEval, Eagle, TechSupportEval

Jan. 2024–Dec. 2024

- Contributed to OpsEval (FSE 2025, CCF A; 3rd author), building a comprehensive benchmark with 9,000+ QA pairs across eight subdomains and multiple evaluation paradigms, plus FAE-Score for expert-aligned automatic evaluation.
- Contributed to Eagle (FSE 2026, CCF A; 4th author) and TechSupportEval (IJCNN 2025, THU B / CCF C; 4th author), studying traceable evaluation-data synthesis from hierarchical documents and constraint-aware automatic evaluation for technical QA.

Research Output

Full publication list: [Google Scholar](#).

- First-author, accepted:** TameR (ICML 2026, CCF A); SPRINT (ICML 2026, CCF A).
- First-author, under review:** IntentAD (ICSE 2027, CCF A); G2AD (AAAI 2027, CCF A).
- First-author, planned submission:** IPD (ICLR 2027).
- LLM/AIOps collaborations:** SIS (planned for ICLR 2027); FoundRoot (ICSE 2026), Eagle (FSE 2026), ChatTS (VLDB 2025), and OpsEval (FSE 2025), all CCF A; TechSupportEval (IJCNN 2025, THU B / CCF C).
- Other collaborations:** MicroScope (KDD 2024, CCF A); Face3DAdv (IJCV 2024, CCF A); AT3D (CVPR 2023 Highlight, CCF A).

Honors and Awards

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| Outstanding Undergraduate Graduate, Department of Computer Science and Technology, Tsinghua University | Jun. 2023 |
| Outstanding Undergraduate Thesis, Tsinghua University | Jun. 2023 |
| Xiaomi Scholarship | Dec. 2022 |
| Comprehensive Excellence Scholarship, Tsinghua University | Dec. 2021, Dec. 2022 |
| Social Work Excellence Scholarship, Tsinghua University | Dec. 2020 |

Skills

- LLM Post-Training:** SFT, GRPO/RLVR, Agentic RL, trajectory synthesis, multi-turn tool use, memory retrieval, active perception
- Multimodal LLMs:** new-modality extension and alignment, time-series modeling, fine-grained localization, active perception, efficient inference, AIOps
- LLM Evaluation:** benchmark and metric design, high-quality evaluation-data construction, automatic evaluation
- Frameworks:** PyTorch, VeRL, vLLM, Open-R1, TRL, Megatron, DeepSpeed, LLaMA-Factory
- Engineering:** Python, C/C++, Java, Kubernetes, Docker, Node.js, Django, MongoDB, MySQL, LaTeX

Profile

- End-to-end LLM research experience spanning problem formulation, data and trajectory construction, model training, evaluation, and deployment validation.
- Strong focus on connecting research novelty with business impact; multiple projects validated or deployed in real enterprise environments.
- Hands-on experience extending training frameworks, adapting inference services, and translating research ideas into working systems.
- Self-driven and collaborative; able to lead research independently and make key contributions to joint projects.